

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: ONE

CLASS: SS3

SUBJECT: Digital Technology

TOPIC: System Development Life Cycle

Introduction

- The System Development Life Cycle (SDLC) is the structured process by which information systems are planned, built and maintained. It turns a problem into a working software solution.

The phases

- **1. Problem definition/identification** — understand the user's problem and objectives.
- **2. Feasibility study** — is it technically, economically, operationally and legally possible? (cost–benefit analysis).
- **3. System analysis** — study the existing system, gather requirements (interviews, questionnaires, observation), produce the system specification.
- **4. System design** — design input, output, process, files and databases; draw diagrams (flowcharts, data flow diagrams, entity-relationship diagrams).
- **5. Implementation** — coding, testing, conversion (parallel, phased, direct or pilot) and installing.
- **6. Maintenance** — correcting errors, adapting to change, perfecting and preventing problems; also documentation and training end users.

Models

- The waterfall model (phases in strict order); the iterative/incremental model; the spiral model (risk-driven); prototypes (a working model shown to users early).

Importance

- The SDLC prevents costly errors, keeps projects on time and within budget, produces quality software and gives clear documentation for future maintenance.

CLASS EVALUATION

1. Define the SDLC.
2. List the six phases in order.
3. What is a feasibility study and why is it done?

4. Differentiate between parallel and phased conversion.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: TWO

CLASS: SS3

SUBJECT: Digital Technology

TOPIC: Data Security & Privacy

Introduction

- Data security is the protection of data against unauthorised access, alteration, destruction and disclosure; privacy is the right of individuals to control what personal information is collected and how it is used.

Threats to data

- **Physical** — theft of devices, fire, flood, power surges, hardware failure. **Human** — careless staff, social engineering (phishing), insider attacks, password sharing. **Technical** — hacking, malware, spyware, ransomware, denial of service, data interception on unsecured networks.

Security measures

- **Access control** — passwords, PINs, biometrics (fingerprints, face), two-factor authentication, access rights by role.
- **Encryption** — scrambling data so only authorised users can read it (HTTPS, file encryption).
- **Firewalls and antivirus** — block intrusions and remove malware; **backups** — regular copies, off-site, tested restores; **updates and patches**; **physical controls** — locks, CCTV, restricted rooms; **policies** — acceptable-use rules, staff training.

Privacy rights and law

- Personal data (names, contacts, health, finances) must be collected lawfully, used only for its purpose, kept accurate and secure, and deleted when no longer needed. Laws: Nigeria's Nigeria Data Protection Act 2023 (and NDPR); the GDPR of the EU; organisations must obtain consent and guard against identity theft.

CLASS EVALUATION

1. Define data security and privacy.
2. List three human threats to data.
3. Explain encryption and why it matters.
4. State two rights of a data subject under Nigerian law.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: THREE

CLASS: SS3

SUBJECT: Digital Technology

TOPIC: Computer Viruses & Protection

Introduction

- A computer virus is a malicious program that attaches itself to other files and replicates, damaging or disrupting the computer. Malware in general includes viruses, worms, Trojans, ransomware, spyware and adware.

Types of malware

- **Virus** — attaches to a host file, spreads when the file runs (boot sector, macro, file viruses). **Worm** — self-replicating, spreads over networks without a host. **Trojan** — hides inside a seemingly useful program. **Ransomware** — encrypts your files and demands payment. **Spyware/keyloggers** — secretly record keystrokes and spy on activity. **Adware** — floods the screen with ads. **Rootkits** — hide deep in the system to control it secretly.

How they spread

- Email attachments and links; infected flash drives and downloads; malicious websites; pirated software; unpatched software vulnerabilities; Bluetooth and network shares; social engineering. Symptoms: slow performance, pop-ups, files disappearing or changing, unexpected messages, unusual network activity.

Protection and control

- Install and update antivirus/anti-malware software; run it regularly; keep the OS and apps patched; avoid unknown attachments and pirated software; back up data; use strong passwords and firewalls; be alert to phishing; educate users; restrict downloads; and disconnect infected machines from the network immediately.

CLASS EVALUATION

1. Define a computer virus.
2. Differentiate between a virus and a worm.
3. State two symptoms of malware infection.
4. List four ways to protect a computer from viruses.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FOUR

CLASS: SS3

SUBJECT: Digital Technology

TOPIC: Emerging Technologies (AI, Cloud Computing)

Introduction

- Emerging technologies are new tools changing how we compute, work and live. Major areas: Artificial Intelligence (AI), cloud computing, the Internet of Things (IoT), blockchain, big data and virtual/augmented reality.

Artificial Intelligence

- AI is the simulation of human intelligence by machines — learning, reasoning, perception and problem-solving. Branches: machine learning (systems learn from data), natural language processing (chatbots, translation), computer vision (face recognition), robotics, expert systems and speech recognition. Applications: assistants (Siri, Alexa), recommendation engines, self-driving cars, medical diagnosis, fraud detection. Benefits: speed, accuracy, automation. Risks: job displacement, bias in decisions, privacy, and misuse.

Cloud computing

- Using remote servers over the internet to store, manage and process data, instead of local machines. Service models: **IaaS** (infrastructure), **PaaS** (platform), **SaaS** (software — Google Workspace, Microsoft 365). Deployment: public, private, hybrid clouds. Benefits: access anywhere, scalability, pay-as-you-go, automatic updates, collaboration. Risks: internet dependency, security and privacy, vendor lock-in, cost at scale.

Other technologies

- IoT** — everyday devices connected to the internet (smart homes, wearables). **Blockchain** — a tamper-proof distributed ledger (cryptocurrencies, supply chains). **Big data** — analysing huge data sets for insights. **VR/AR** — immersive training and games. **5G** — faster mobile broadband enabling these technologies.

CLASS EVALUATION

- Define AI and name three of its branches.
- State two benefits and two risks of AI.
- Define cloud computing and name its three service models.

4. What is the IoT? Give two examples.

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FIVE

CLASS: SS3

SUBJECT: Digital Technology

TOPIC: Computer Applications (Revision & Application)

Introduction

- This week revises how computers are applied across sectors, and the practical skills needed for the WASSCE/NECO computer studies paper: word processing, spreadsheets, presentations and the internet.

Applications of computers

- **Education** — e-learning, CBT exams, educational software. **Banking** — ATMs, electronic transfers, mobile banking, POS. **Medicine** — patient records, diagnosis imaging, telemedicine. **Business** — accounting, payroll, e-commerce, inventory. **Government** — e-governance, INEC voter register, NIN enrolment. **Engineering** — CAD designs, simulations. **Communication** — email, social media, video conferencing. **Agriculture** — farm records, drones, market information.

Practical skills

- Word processing: create, edit, format, tables, headers/footers, spellcheck, mail merge, print documents (MS Word, Google Docs). Spreadsheets: enter data, formulas (=SUM, =AVERAGE), functions, charts, sorting/filtering (Excel, Google Sheets). Presentations: slides, themes, transitions, animations, speaker notes (PowerPoint). Internet: search, email, web pages, e-safety, file sharing, web conferencing.

CLASS EVALUATION

1. State two applications of computers in banking and education.
2. Give the spreadsheet formula to add cells A1 to A10.
3. What is mail merge?
4. List three e-safety rules for social media.